

Applications of Mixture Discriminant analysis

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Abstract

Mixture Discriminant Analysis (MDA) is a classifier algorithm that extends traditional Gaussian Discriminant Analysis (GDA) by incorporating more flexibility using mixture models. Unlike GDA, which assumes a single Gaussian distribution per class, MDA models each class as a mixture of multiple parametric components, allowing for the representation of sparse and multi-clustered distributions within class labels. This approach also enhances the ability to classify data with overlapping distributions. MDA combines unsupervised clustering within classes with discriminant analysis, optimizing both the fit of the mixture models and the separation between classes. This study reviews the principles of MDA, highlights its computational framework, and discusses its advantages over traditional discriminant techniques. Additionally, applications and challenges, including model selection, are addressed, showcasing MDA characteristics for classification in diverse domains.

Keywords

Expectation-Maximization - Gaussian mixture models - Discriminant analysis

1 Introduction

Among unsupervised machine learning approaches, k -means [1] stands out as one of the most natural way to separate data into different classes. However, by applying k -means, one adds strong hypotheses to the data distribution that may not always hold in real situation. For example, differentiating classes with same means but different variances would not give convincing results with k -means. As such, Mixture Models can be viewed as probabilistic extensions for unsupervised learning problems, in which points are considered to be drawn according to one among several distributions, each with different parameters [2]. Those can thus adapt to more complex distributions (for example if two classes behave differently due to a latent unknown variable). A particular case of Mixture models is Gaussian Mixture Models (GMMs) that suppose Gaussian distribution for each class. There are several applications of mixture models that range from speaker recognition [3] to PET imagery [4].

In this work, we will present the theoretical foundations of GMMs. We will explain the evidence lower bound (ELBO)[5] approximation that will allow us to optimize Mixture Models in general via the EM algorithm. In the particular case of GMMs, we will detail the optimization for the classes proportions and for the Gaussian parameters. From GMMs, we derive classification algorithms by assigning each point to the class maximizing its likelihood under the mixture model. [6] [7]. We then apply an algorithm called Gaussian Discriminant Analysis (GDA) to concrete cases to show strengths

and weaknesses of the method [8].

2 Theoretical Background

This section introduces the foundational concepts underlying Gaussian Mixture Discriminant Analysis (GMDA): Gaussian Mixture Models and Discriminant Analysis. We will first discuss the ELBO framework and the Expectation-Maximization (EM) algorithm, which are key tools for parameter estimation in Gaussian Mixture Models.

2.1 ELBO trick

We consider n observations modeled by a vector X of random variables. We suppose that those observations are explainable from a set of P latent variables Z . From Z distribution, we get X using $p(X|Z)$. We drawn the graphical model in Fig.1

We make assumptions on the distribution Z , to get a model for $p(Z)$.

Using Bayes and total probability laws, we get:

$$p(Z|X) = \frac{p(X|Z)p(Z)}{p(X)},$$

and the marginal distribution along the observed variables:

$$p(X) = \int p(X|Z)p(Z)dZ.$$

To find our latent variable distribution, we need to estimate the posterior probability $p(Z|X)$, that requires the computation of $p(X)$ which can possibly have no closed form or can be hard to compute in our problems.

The ELBO trick will link the distribution of X , $p(X)$, to a lower bound defined from a surrogate function $U(Z) \simeq p(Z|X)$.

Let us derive the formulas in the most generic case.

Let (X, μ) be a measured space, we suppose that $p(Z|X)$ is absolutely continuous with respect to μ . We will use the same notation for the density and the probability measure $p(Z|X)$. We want U to approximate the density function of $p(Z|X)$ i.e. We will want $p(Z|X) \simeq U$. We can use the Kullback-Leibler divergence, denoted as D_{KL} , to measure how close those two density functions are:

$$D_{KL}(U||p(Z|X)) = \int_X U(z) \log \left(\frac{U(z)}{p(Z|X)(z)} \right) d\mu(z)$$

If we suppose that $\mathbb{E}_U[\log(U)]$ and $\mathbb{E}_U[\log(p(Z|X))]$ are finite, we get:

$$\begin{aligned} D_{KL}(U||p(Z|X)) &= \int_X U(z) \log(U(z)) d\mu(z) \\ &\quad - \int_X U(z) \log(p(Z|X)(z)) d\mu(z) \end{aligned}$$

We suppose that $\mathbb{E}_U[\log(p(Z, X))]$ is finite:

$$\begin{aligned} D_{KL}(U||p(Z|X)) &= \mathbb{E}_U[\log(U)] - \mathbb{E}_U[\log(p(Z|X))] \\ &= \mathbb{E}_U[\log(U)] + \mathbb{E}_U[\log(p(X))] - \mathbb{E}_U[\log(p(Z, X))] \\ &= \mathbb{E}_U[\log(U)] + \log(p(X)) - \mathbb{E}_U[\log(p(Z, X))] \end{aligned}$$

If we define the entropy as: $\mathbb{H}(U) = -\mathbb{E}_U[\log(U)]$, we get:

$$\log(p(X)) = D_{KL}(U||p(Z|X)) + \mathbb{H}(U) + \mathbb{E}_U[\log(p(Z, X))]$$

$D_{KL}(U||p(Z|X))$ is intractable, $\mathbb{H}(U)$ and $\mathbb{E}_U[\log(p(Z, X))]$ can be computed for a given U and X models.

As $D_{KL}(U||p(Z|X)) \geq 0$ and null if U and $p(Z|X)$ are equal almost everywhere:

$$\log(p(X)) \geq \mathbb{H}(U) + \mathbb{E}_U[\log(p(Z, X))]$$

The ELBO is defined as the quantity: $\mathbb{H}(U) + \mathbb{E}_U[\log(p(Z, X))]$.

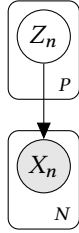


Figure 1: A graphical model depicting a latent variable model with N data points and P latent variables

2.2 EM algorithm

We suppose that X is parametrized by some parameters $\theta_X \in \mathbb{R}^d$ and Z the latent variables are also parametrized by $\theta_Z \in \mathbb{R}^{d'}$. Our graphical model is presented in Fig.2.

The EM algorithm is based on the ELBO trick to find approximation for θ_X and θ_Z given the n observations. It should find approximation which maximizes locally the log-likelihood.

Algorithm 1 Expectation-Maximization

Initialize θ_X, θ_Z

while not converged **do**

E STEP: based on the current θ_Z, θ_X ,

 Compute $U := p(Z|X; \theta_X, \theta_Z)$.

M STEP: find θ_X, θ_Z which maximise the ELBO:

$$\theta_X, \theta_Z = \arg \max_{\tilde{\theta}_X, \tilde{\theta}_Z \in \mathbb{R}^d \times \mathbb{R}^{d'}} \mathbb{H}(U) + \mathbb{E}_U[\log(p(Z, X; \tilde{\theta}_X, \tilde{\theta}_Z))]$$

end while

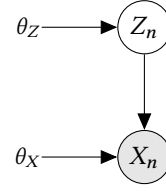


Figure 2: A graphical model depicting a latent variable model with N data points parametrized by θ_X and latent variables parametrized by θ_Z .

2.3 Gaussian Mixture Models (GMM)

A simple model that relies on the previously defined EM algorithm to fit its coefficients is the Gaussian Mixture Model.

First, let us consider a set of random variables $(X_i)_{1 \leq i \leq n}$, and let us assume that $X_i \sim \sum_{k=1}^K \pi_k \mathcal{N}(\mu_k, \Sigma_k)$, $\mathcal{N}(\mu_k, \Sigma_k)$ being the Gaussian distribution of parameters (μ_k, Σ_k) , where $K \in \mathbb{N}^*$ and the priors $\pi_k \geq 0$ for $1 \leq k \leq K$ and $\sum_{k=1}^K \pi_k = 1$. We note $\theta_k = (\mu_k, \Sigma_k)$, the parameters of the k^{th} cluster of the mixture model. The graphical model is represented in Fig. 3.

Given $(x_i)_i$ a realization of $(X_i)_i$, the likelihood of a model of K Gaussian components parameterized by π and θ thus writes like so:

$$\mathcal{L}_{GMM}(\pi, \theta | x) = \prod_{i=1}^n \sum_{k=1}^K \pi_k \mathcal{N}(x_i | \mu_k, \Sigma_k)$$

To help estimate π and θ , we can introduce the auxiliary variables $(Z_i)_i$ where for each index i , $z_i \in \{0, 1\}^K$ and $z_{i,k} = 1$ if and only if x_i is drawn according to the k^{th} component. The main objective of a Gaussian mixture model is to estimate the values of the latent variables $(z_i)_i$ based on the observations $(x_i)_i$. Given, $Z_i \sim \text{Mu}(1, \pi)$, with $\text{Mu}(1, \pi)$ being the multinomial distribution with probability vector π , and $(X_i | Z_{ik} = 1) \sim \mathcal{N}(\mu_k, \Sigma_k)$.

To define an appropriate clustering, we want to find for each data point the most likely component from which it was drawn. For this model we can write the complete log-likelihood:

$$\begin{aligned} \ell_C(\pi, \theta | \mathbf{x}, \mathbf{z}) &= \log \prod_{i=1}^n \prod_{k=1}^K [p(z_{i,k} = 1) p(x_i | z_{i,k} = 1, \pi_k, \theta_k)]^{z_{ik}} \\ &= \log \prod_{i=1}^n \prod_{k=1}^K [\pi_k \mathcal{N}(x_i | \mu_k, \Sigma_k)]^{z_{ik}} \\ &= \sum_{i=1}^n \sum_{k=1}^K z_{i,k} \log(\pi_k \mathcal{N}(x_i | \mu_k, \Sigma_k)) \end{aligned} \quad (1)$$

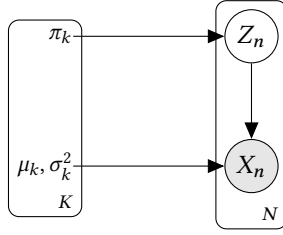


Figure 3: A graphical model depicting a latent variable model with N data points modeled by gaussian and K mixture components.

2.3.1 Application of EM algorithm in GMM context. In this section, we will describe the EM algorithm in the case of a GMM setting. The EM algorithm is based on two steps detailed in the following sections. We iterate this two steps until convergence. In the next sections, we will denote the iterations as t .

2.3.2 E step.

At step t , we introduce the quantity:

$$Q(\theta | \theta^{(t)}) = \mathbb{E}_{Z \sim p(\cdot | \mathbf{x}, \pi, \theta^{(t)})} [\ell_C(\pi, \theta | \mathbf{x}, \mathbf{z})]$$

With the analytical form $p(z_i | x_i, \pi, \theta) = \text{Mu}(z_i | 1, \tau_i)$ and

$$\tau_{i,k} = \frac{\pi_k \mathcal{N}(x_i | \mu_k, \Sigma_k)}{\sum_{l=1}^K \pi_l \mathcal{N}(x_i | \mu_l, \Sigma_l)}$$

the probability of i belonging to the k^{th} component given x_i . By (1), we obtain:

$$Q(\theta | \theta^{(t)}) = \sum_{i=1}^n \sum_{k=1}^K \tau_{i,k}^{(t)} \log \left[\pi_k^{(t)} \mathcal{N}(x_i | \mu_k^{(t)}, \Sigma_k^{(t)}) \right] \quad (2)$$

2.3.3 M step.

We then maximize Q and get the next $\theta^{(t+1)}$:

- (1) $n_k := \sum_{i=1}^n \tau_{i,k}^{(t)}$
- (2) $\pi_k^{(t+1)} = \frac{n_k}{n}$
- (3) $\mu_k^{(t+1)} = \frac{\sum_{i=1}^n \tau_{i,k}^{(t)} x_i}{n_k}$
- (4) $\Sigma_k^{(t+1)} = \frac{\sum_{i=1}^n \tau_{i,k}^{(t)} (x_i - \mu_k^{(t+1)}) (x_i - \mu_k^{(t+1)})^T}{n_k}$

Algorithm 2 Expectation-Maximization for a Gaussian Mixture Model

Initialize $K \in \mathbb{N}^*$ and the clusters parameters $(\pi_k^{(0)})_k, (\theta_k^{(0)})_k$ with a simple clustering algorithm.

while not converged **do**

E: Compute $(\tau_{i,k}^{(t)})_k$ using $(\pi_k^{(t)})_k, (\theta_k^{(t)})_k$ for all $i \in \llbracket 1, n \rrbracket$.

M: Compute the new estimated $(\pi_k^{(t+1)})_k, (\theta_k^{(t+1)})_k$ based on

$(\tau_{i,k}^{(t)})_k$.

end while

We note that the initialization step is usually done using the k -means algorithm. Then, based on the clusters, we fit the initial parameters of the algorithm:

$$(\theta_k^{(0)})_k = (\mu_k^{(0)}, \Sigma_k^{(0)})_k.$$

2.4 Student Mixture Model (tMM)

A simple extension of the previous mixture model is to consider other parametric distribution. For instance, we can replace the Gaussian distribution by the Student's t-distribution for more robust density estimation. The mixture model is then:

$$X_i \sim \sum_{k=1}^K \pi_k \int \mathcal{N}(\mu_k, \Sigma_k) \text{Ga}\left(z_i | \frac{v_k}{2}, \frac{v_k}{2}\right) dz_i,$$

where $\text{Ga}(\cdot | \frac{v_k}{2}, \frac{v_k}{2})$ is the Gamma probability density function of parameters $(\frac{v_k}{2}, \frac{v_k}{2})$. The particularity of this model is the added parameter v_k for all $1 \leq k \leq K$ that is called the *degree of freedom* of the k^{th} Student's distribution. In this study, we will consider the values of v_k are known beforehand for all $1 \leq k \leq K$, it is then easy to fit the Gaussian parameters using a EM algorithm knowing the values of $(v_k)_k$.

2.5 Discriminant analysis

2.5.1 General case.

Discriminant analysis (DA) is a generative classifier method. Given a vector $\mathbf{x}$, the aim is to determine the label $y \in C$, where C is the label space, associated with the input $\mathbf{x}$ using probabilistic arguments. The core idea in DA is to consider parameterized conditional probability distributions $p(\mathbf{x} | y = c, \theta)$ given a class $c \in C$ for the target variable, and θ belonging to a parameter space. The most universal method for discriminant analysis is defined using Gaussian likelihood:

$$p(\mathbf{x} | y = c, \theta) = \mathcal{N}(\mathbf{x} | \mu_c, \Sigma_c).$$

Using Bayes formula, the goal is to determine the class that maximizes the posterior distribution:

$$\begin{aligned} \hat{y}(\mathbf{x}) &= \arg \max_{c \in C} \log(p(y = c | \mathbf{x}, \theta)), \\ &= \arg \max_{c \in C} \log(\pi_c) + \log(p(\mathbf{x} | \mu_c, \Sigma_c)), \end{aligned} \quad (3)$$

with π_c being the prior applied to a class c .

In the Gaussian setting, the right term is actually a negative Mahalanobis distance between $\mathbf{x}$ and μ_c . The problem can be interpreted as an optimization problem of the distance between $|C|$ cluster centroids [9].

Studying the boundaries between each class is also a point of interest, as it is where the predictions can change dramatically from one model to another. Indeed the boundary equation between two classes $c, c' \in C$ is defined as

$$p(y = c | \mathbf{x}, \theta) = p(y = c' | \mathbf{x}, \theta). \quad (4)$$

In a hard classification setting, the choice of the more likely class is done based on the mode of the probability distribution. Dealing

with the region where the probabilities between two classes are really similar is essential to determine an adequate predicted label.

2.5.2 Linear discriminant analysis (LDA).

To evaluate the efficiency of a Gaussian mixture-based discriminant analysis, we need to introduce a simpler approach as a benchmark. The linear discriminant analysis is a subclass of the conditional generative classifier problems. Indeed, we consider the covariance matrices to be tied in between classes, this means that for all $c \in C$, $\Sigma_c = \Sigma$, where Σ is constant for all classes. The efficiency of the model is due to its simplicity, only one covariance matrix Σ needs to be fitted on the data.

If we develop the quadratic term inside the Normal distribution, we can actually find the expression of the log posterior distribution:

$$\begin{aligned} \log(p(y = c|\mathbf{x}, \theta)) &\propto \log(\pi_c) + \boldsymbol{\mu}_c^T \Sigma^{-1} \mathbf{x} - \frac{1}{2} \mathbf{x}^T \Sigma^{-1} \mathbf{x} \\ &\quad - \frac{1}{2} \boldsymbol{\mu}_c^T \Sigma^{-1} \boldsymbol{\mu}_c, \\ &\propto \log(\pi_c) - \frac{1}{2} \boldsymbol{\mu}_c^T \Sigma^{-1} \boldsymbol{\mu}_c + \boldsymbol{\mu}_c^T \Sigma^{-1} \mathbf{x}. \end{aligned} \quad (5)$$

The quadratic term $-\frac{1}{2} \mathbf{x}^T \Sigma^{-1} \mathbf{x}$ vanishes with the normalization. We will note the constants $\alpha_c = \log(\pi_c) - \frac{1}{2} \boldsymbol{\mu}_c^T \Sigma^{-1} \boldsymbol{\mu}_c$ and $\boldsymbol{\beta}_c = \Sigma^{-1} \boldsymbol{\mu}_c$. So the log probability becomes proportional to $\alpha_c + \boldsymbol{\beta}_c^T \mathbf{x}$.

The equation (5) yields a linear boundary between classes. Substituting the result from (5), the results become:

$$\alpha_c - \alpha_{c'} = (\boldsymbol{\beta}_c - \boldsymbol{\beta}_{c'})^T \mathbf{x}. \quad (6)$$

The boundary between classes is thus linear.

2.5.3 Gaussian mixture discriminant analysis (GMDA).

In this study, we will focus more on the following model. Instead of assigning each class a unique Gaussian density, we use a mixture of Gaussian distributions. This model is named Gaussian mixture discriminant analysis (GMDA) [8]:

$$p(\mathbf{x}|y = c, \theta) = \sum_{i_c=1}^{k_c} \pi_{i_c} \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_{i_c}, \Sigma_{i_c}). \quad (7)$$

The idea behind a mixture DA model is to use the properties of a GMM with k_c components for each class c in C . The attribution of a class label would then be determined by the proximity between the different clusters with the input $\mathbf{x}'$. To implement such model, we need to fit a GMM model on each class subsample.

Different methods can be tested based on the adequate number of components. A simple approach could be to consider a constant value for $k \in \mathbb{N}^*$, such that $(k_c)_{c \in C} = (k)_{c \in C}$, but more elaborate ones would require finding the number of components with the greatest information criterion.

Based on what precedes, we can define analogously a Student's mixture discriminant analysis using tMM.

2.6 Model selection

Model selection in the context of Gaussian mixture models is usually achieved by maximizing a penalized log-likelihood criterion, the

two most common ones are the Akaike's information criterion (AIC) [10] and the Bayesian information criterion (BIC) [11]:

$$BIC = \log(\ell_C) - \frac{d_K \log(n)}{2}. \quad (8)$$

$$AIC = \log(\ell_C) - d_K. \quad (9)$$

In this context d_K is the number of the parameters of the model and n is the dataset length.

For a simple K component Gaussian mixture model in $\mathbb{R}^d$,

$$d_K = Kd + K \frac{(d+1)d}{2}. \quad (10)$$

The added penalty term in $\log(n)$ makes the BIC usually more resilient to the dataset size. It tends to better avoid over-fitting the optimal value of K over the original dataset than the AIC. Despite their simple implementation, in the context of mixture discriminant analysis, these methods fail to capture the entirety of the penalized problem [12], instead of penalizing the classification results, only the mixtures are penalized. Model selection in the context of discriminant analysis relies of l -fold cross-validation error rates. For several folds of the training sets, we compute the error rate and then determine the model which minimizes the averaged error rate.

Toy experiments.

To evaluate how GMDA compares with LDA, we led several experiments. We compared our implementation of GDA with a simple LDA classifier.

We deliberately chose to generate non-linearly separable data point clouds to best show LDA limitations. In a first toy experiment, we fitted a GMM model on each class, with 8 mixtures per class. The results are shown in Fig. 4a, we clearly see the struggle of LDA, which achieves an accuracy of 0.42, compared to an accuracy of 0.90 for the mixture model. Visually, we can observe that the ellipses representing LDA model covariances do not fit properly the synthesized data. As we have mentioned earlier, the covariance matrix is shared between classes, meaning that for sparse and uneven distributions, the fitted LDA will yield underwhelming results. However, we can observe over-fitting of the GMM model that uses too many gaussians to represent the underlying distribution, which is an undesirable property.

To study the impact of log-likelihood criterion to optimize the hyperparameters of the model, we use BIC model selection. We set the max number of Gaussian distributions to 8 for each class. The results are presented in Fig. 4b. The number of Gaussian models used to fit the distributions is reduced, and the accuracy remained to .90, which shows the benefits of using the BIC criterion on some instances.

3 Experiments

In this section, we aim to provide examples of experiments on real datasets. The underlying objective of each study is to determine the efficiency of the model for classification tasks. We propose the following studies:

- Given a dataset of images of eye vessels, we aim to determine whether a pixel is a blood vessel or not. The aim is to do binary classification using the GMDA model.
- Given a survey of gene expressions, we aim to determine whether a patient has Alzheimer’s disease.
- We study the likeliness of the use of GMDA in a few-shot classifying setting, i.e. with few training samples, notably in “fine-tuning” tasks in natural language processing.

3.1 Segmentation experiment

In this section, we explore the application of the Gaussian Mixture Discriminant Analysis (GMDA) framework to the task of segmenting blood vessel structures from background regions in eye images. The core aim is to classify pixels (that will be represented as vector of $\mathbb{R}^3$) as either part of the blood vessels or as belonging to the background. A key focus of this investigation is understanding how the number of Gaussian components in the mixture impacts the framework’s ability to perform accurate segmentation.

As shown in Figure 5, GMDA distinguishes well bigger blood vessel while struggling with the smaller ones. Complementing this, Figure 7 highlights the learned marginal distribution for the blue channel of the input images. The model encounters challenges in managing sharp spikes and asymmetric transformations within this channel, a difficulty that extends to other color channels as well. To delve deeper into the role of the number of Gaussian components, Figure 6 presents segmentation outcomes for mixtures containing between 1 and 15 components.

Finally, recognizing the limitations of pixel-wise predictions, we explored the use of patch-based predictions to leverage the spatial coherence of image regions. This approach aims to capture more contextual information, potentially enhancing the consistency and accuracy of segmentation. Figure 8 illustrates how larger patches contribute to more stable and coherent segmentation results, which reduces noise and improves the detection of smaller blood vessel.

3.2 Alzheimer’s disease study

Based on the dataset provided by the study [13], the aim is to evaluate the impact of $G = 8,650$ genes for a pathological diagnostic of Alzheimer disease over a control group of 180 people and a test group of 176 cases. In this section, we apply the GMDA method to confirm the predictability of the disease based on the genes. The input $\mathbf{x}$ is a vector of gene expressions that lies in $\mathbb{R}^G$ for one patient. The output data $\mathbf{y}$ is a binary classification of whether signs of the pathology have manifested or not.

Due to the large number of genes tested, a step of dimension reduction is required beforehand to limit the computational time of the algorithm and the large uncertainty due to the high number of parameters to be fit.

We consider the problem under the model selection setting, the aim is to determine hyperparameters that fit better the problem. Our implementation is the following: first, we apply a simple dimension reduction algorithm such as a principal components analysis (PCA), we then apply a GMDA model for binary classification. The parameters to be determined are the number of components as well

as the number of Gaussian distribution considered for both classes (we will take the same number of mixtures for both classes).

The results are illustrated on the heatmap on Fig. 9. For each test set on a 4-fold cross-validation, we compute the different accuracies as minus the error rate (we want to maximize the accuracy), and then average them. We then plot the averaged value on a heatmap given the number of principal components and the number of components for each classes. The best results are achieved for low number of GMM components but relatively large number of PCA components.

What is surprising is the presence of an “inaccurate zone” shown on the heatmap. This is likely to be an over-fit phenomena in a low-dimension setting while having a large number of components. This suggests that the choice of PCA components plays a critical role in balancing variance retention and overfitting risks.

When the number of PCA components is low, the GMM primarily relies on the most relevant features that explain the majority of the variance, reducing the risk of overfitting to less significant dimensions. Conversely, with a high number of PCA components, the PCA retains the full data structure, minimizing the risk of excluding important information. However, in intermediate cases, certain combinations of PCA components may lead to overfitting by the GMM on less relevant dimensions. This highlights a key limitation of PCA: it selects components based only on variance explanation, without considering their relevance for the specific classification task.

3.3 Few-shot classifier

In this section, we aim to check the performances of the GMDA classifier in the context of few-shot learning. We want to test the efficiency of the model on text-related learning task.

The methodology of the experiment is the following, we consider the Yelp reviews dataset, with the input data being online reviews and the label being the rating associated with it. We first take a generic pre-trained transformer encoder model such as `msmarco-distilbert-dot-v5` to embed the sentences into dense representation spaces (for the model mentioned the dimension is $d = 768$). We then fit a PCA model to reduce the dimension down to $d' = 5$, this is done to compensate the lack of available data for training. We apply the GMDA classifier on the dense representation and compare it with a simple ReLU neural network. The accuracy of the GMDA is not as good as the one of the neural network as illustrated in Fig. 10.

One reason for GMMs under-performance can be that, as [14] showed, BERT embedding do not follow Gaussian distributions, as words of high-frequency are densely close to the origin while those of low-frequency are sparsely distributed far from the origin. The paper then apply normalizing flows to transform the BERT embedding into a smooth isotropic Gaussian distribution. Moreover, performances limitations of GMMs can also be explained by the application of PCA, that assumes that the most informative features can be captured by linear combination of original features.

This may not always hold and may cause information loss, by overlooking non-linear patterns, when lowering the dimension via this method. Furthermore, discriminant analysis using tMMs yielded similar results to the GMM-based model.

4 Limits of the mixture discriminant analysis

In this section, we highlight the model’s flaw and the challenges faced when implementing it.

4.1 Over-fitting tendencies of mixture discriminant analysis

One major flaw of the mixture discriminant analysis is highlighted by Fig. 9. In practice, for many applications, the evaluation accuracy is greater for models using only one Gaussian blob by class. This shows that the model have a tendency to overfit when using several Gaussian blobs. An example of such behaviour is observed on the left image of Fig. 4a on which too many blobs are fitted on class 2, whereas only two blobs were used when generating the data. In general, to avoid these problem, a selection criterion as seen in 2.6 can be used to limit overfitting on the train set.

4.2 Curse of dimensionality

Another challenge faced by the mixture discriminant analysis is the curse of dimensionality. Indeed, if we set our study in $\mathbb{R}^d$, and we consider the set of class labels C the number of parameters for a mixture model with k mixtures that have to be fitted is in $O(|C|d^2)$ given (10) ($|C|$ being the cardinal of the class set). This can make the method impractical in high-dimensional spaces that would require a substantial training sample size m in order for the model to learn all its parameters. Practical solutions to bypass this problem exist, one can consider a problem setting with tied covariance matrices, where every blob of a same class share the same covariance matrix. Nonetheless, these solutions are usually not useful when dealing with high-dimensional setting in which $d > m$, thus practical solutions to reduce the dimension of the input vectors are often required, this obviously leads to some information loss.

Another issue related to the curse of dimensionality has to do with the parametric properties of the likelihood using Gaussian distributions. One problem is the flattening of the Gaussian distribution in high-dimension, at the mode, the probability density function of a unit Gaussian distribution is $(2\pi)^{-\frac{d}{2}} \xrightarrow{d \rightarrow \infty} 0$. This can lead to more uncertainty when determining a label. Furthermore, in high dimension a "soap bubble" effect can be observed [15]. Marginal distributions of $k < d$ variables will be localized in a restricted subset of the space on which are defined the k random variables.

4.3 Limited expressive power

One key assumption of a parametric density-based discriminant analysis is that they suppose the modelled distribution can be expressed as a mixture of the parametric distribution. This represents a huge prior that may not be suitable for all types of data. For example if we want to fit skewed distributions or fat-tailed distributions,

some methods using user-defined distributions may not be suited, and other mixture models may be required.

This is particularly the case for medical data, that represent extreme events like Alzheimer’s disease, and thus may be fat-tailed. This might violate the normality assumptions of the model and impact its performances. Furthermore, when we leave the Gaussian setting and use another parametric mixture, it results being difficult numerically as it is the case with the Student model, when the degree of freedom is unknown.

The parametric nature of the mixtures can be seen as an advantage for explainability but also as a drawback when it involves modeling in higher dimension. For example, in the few-shot learning experiment, the text data may not exhibit Gaussian distribution behavior, with sparse and skewed word distributions, living in a very high dimension compared to the number of parameters used to describe the model.

5 Conclusion

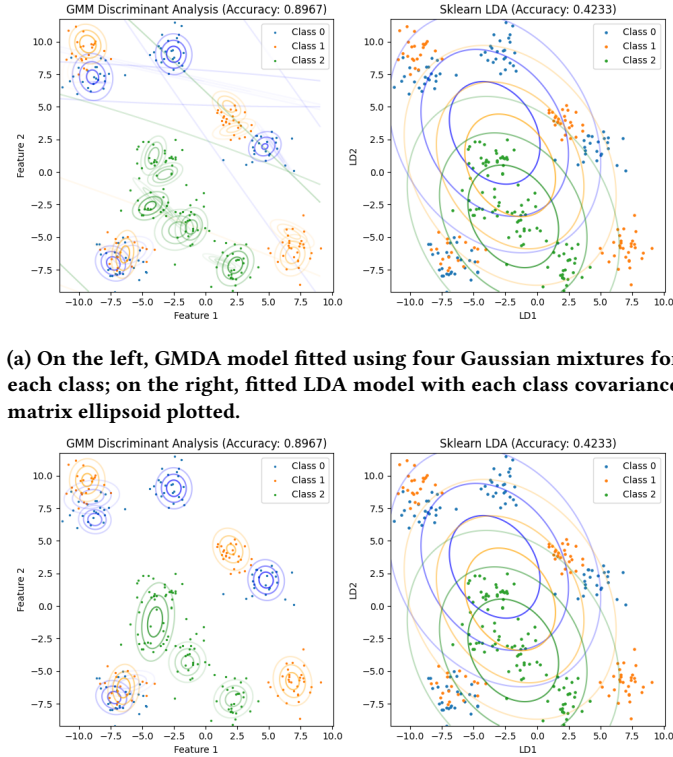
Mixture models are versatile and powerful tools that can be used for both supervised and unsupervised tasks, and can find applications across various domains such as clustering, segmentation, or distribution approximation. They are particularly well-suited for Discriminant Analysis, offering a flexible framework for representing and learning data in a way that can capture complex patterns. The adaptability and mathematical foundation of Mixture models make them a modern data analysis tool for machine learning applications.

However, as we have highlighted in the experiments and discussed in last section, the model suffers from common challenges encountered in machine learning. Indeed, in high dimension its performances can be significantly affected the curse of dimensionality. Even though some practical solutions to bypass these problems do exist on given settings, there are no universal solutions to these issues, and their full potential can only be unlocked if applied on adapted data distributions.

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Appendix: Toy Examples & Results of Eye Vessel Study



(a) On the left, GMDA model fitted using four Gaussian mixtures for each class; on the right, fitted LDA model with each class covariance matrix ellipsoid plotted.

(b) Same experiment as Fig. 4a, where the GMM parameters were selected via the BIC criterion.

Figure 4: Toy examples for mixture discriminant analysis models.

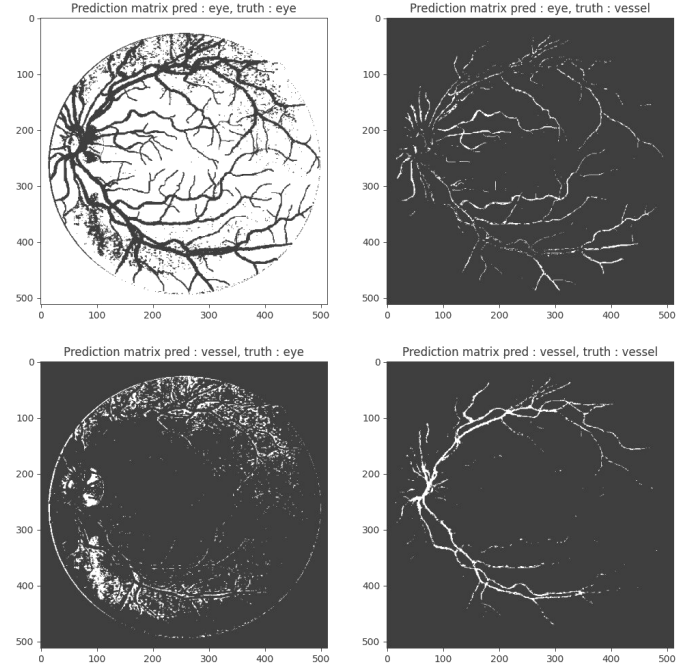
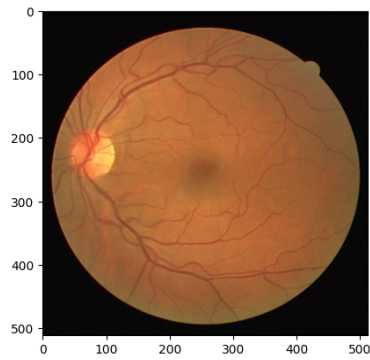
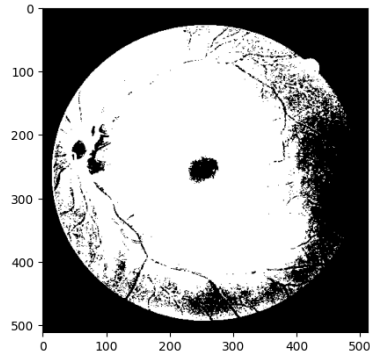


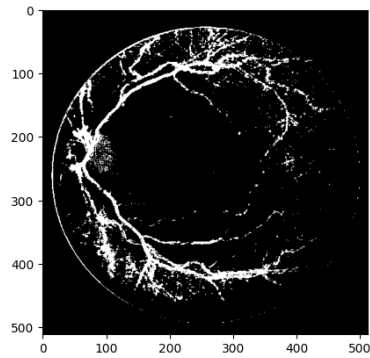
Figure 5: Confusion matrix visualizations for Mixture Discriminant Analysis (MDA) applied to eye vessel segmentation. The top-left quadrant shows adequate classifications of eye background (non-vessel areas). The top-right quadrant highlights the vessel regions misclassified as eye background, sparse distribution suggests a difficulty in capturing fine vessel details. The bottom-left quadrant represents false positives, where eye regions were misclassified as vessels, indicating over-sensitivity in certain areas. The bottom-right quadrant shows correct vessel predictions. However, slight discontinuities suggest challenges in detecting thin or fragmented vessels.



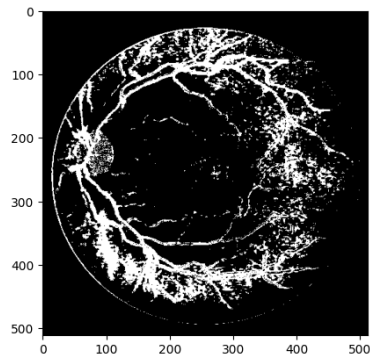
(a)



(b)

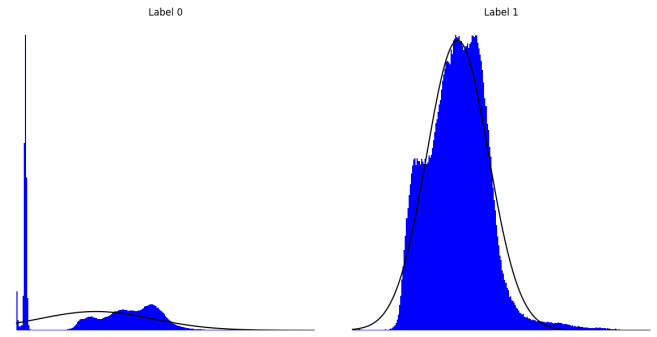


(c)

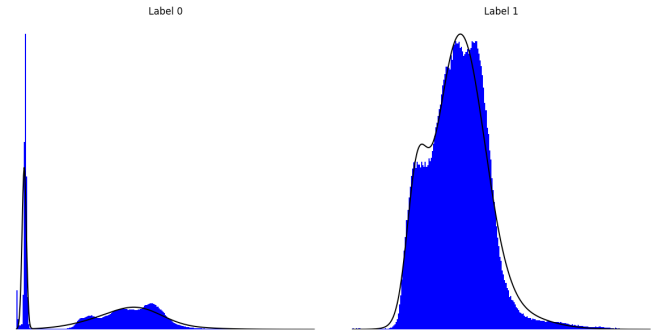


(d)

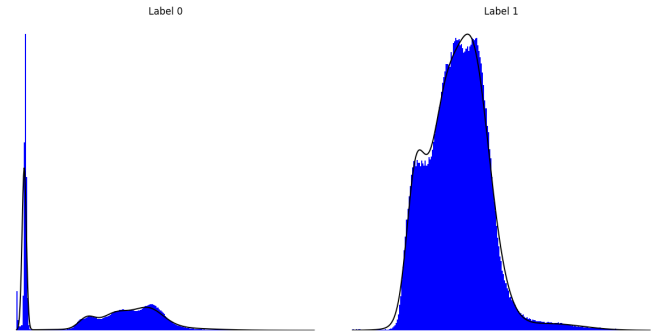
Figure 6: Illustration of MDA results for blood vessel segmentation in eye images showcasing: (a) the original eye image, (b) segmentation using GMM with 1 component, (c) segmentation with GMM using 5 components, and (d) segmentation with GMM using 15 components. The plot highlights the impact of the varying number of components on segmentation details.



(a) GMM with 1 component.



(b) GMM with 5 components.



(c) GMM with 15 components.

Figure 7: Illustration of Gaussian Mixture Model (GMM) blue channel (in RGB) marginal distribution for blood vessel segmentation. (a) shows that one Gaussian is not enough to understand the color distribution of pixels. (b) and (c) show that with 5 or 15 distributions the result are close to the same, the models have a hard time learning the precise distribution.

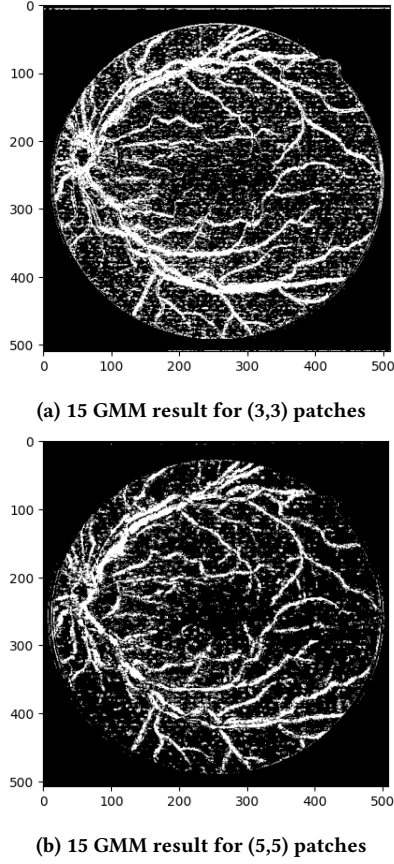


Figure 8: Gaussian Discriminant Analysis results for blood vessel segmentation using different patch sizes

Appendix: Alzheimer Classifier

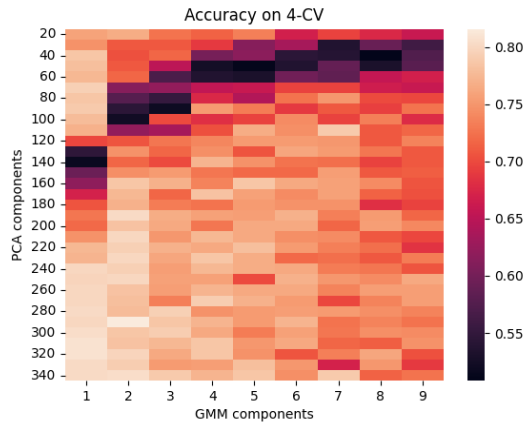


Figure 9: Heatmap of the average accuracy for a 4-fold cross-validation by PCA components and the GMM components.

Appendix: Few-Shot Classifier

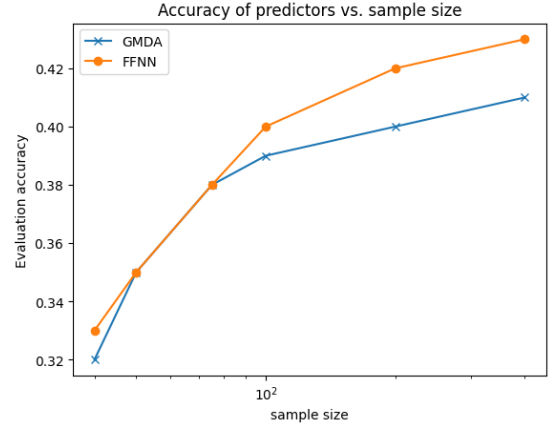


Figure 10: Comparison of accuracy achieved by the feed-forward network (FFNN) and the GMDA model in function of the number of samples in the training set. The FFNN achieves overall better accuracy.

Appendix: Contributions Statement

Kerrian Le Caillec:

- Introduction: 1
- Theoretical background: 2.4, 2.5, 2.6
- Experiments: 3.2, 3.3
- Limits of MDA: 4

Paul Tabbara:

- Theoretical background: 2.3, 2.6.4
- Limits of MDA: 4.3
- EM and BIC implementations

Antoine Vialle:

- Theoretical background: 2.1, 2.2
- Experiment: 3.1
- Experiments with synthetic datasets.